

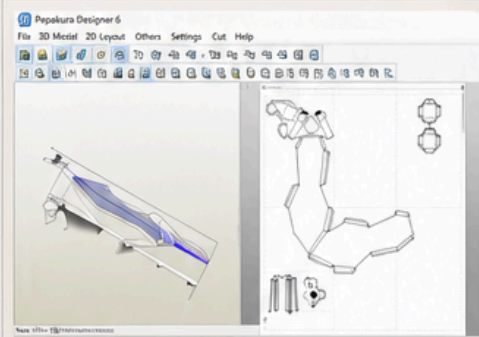
DESIGNING A MINIATURE SWORD

FROM CAD MODEL TO CLAY SCULPTURE



1 DIGITAL CONCEPT

Initial CAD model used to establish overall proportions, silhouette, and geometric features.



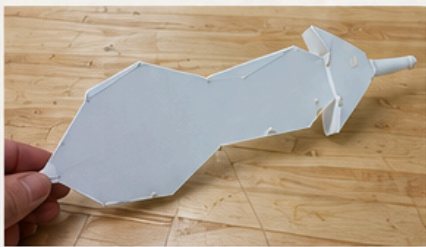
2 PEPAKURA DEVELOPMENT

The CAD model was unfolded into a printable pattern using Pepakura Designer.



3 PAPER ARMATURE CONSTRUCTION

Printed parts were cut, folded, and glued together to create a lightweight internal structure.



4 REINFORCEMENT & SHAPING

The paper armature was reinforced and filled to create a rigid core capable of supporting clay.



5 CLAY SCULPTING

Clay was applied over the armature and sculpted to refine the blade profile, guard, and surface details.



MATERIALS

- | | |
|-------------------|-----------------|
| CAD Software | Foam Filler |
| Pepakura Designer | Clay |
| Cardstock / Paper | Sculpting Tools |
| Adhesive | |

GOALS

- Create a lightweight structure
- Maintain symmetry
- Reduce clay usage
- Establish accurate proportions before sculpting



KEY TAKEAWAY

Using a digital model and paper armature allowed complex shapes to be created quickly while providing a strong foundation for clay sculpting.



CAD MODEL



UNFOLD



PAPER ARMATURE



REINFORCE



CLAY SCULPT